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Expression of Depression Among Arab Twitter Users Using Arabic Corpus Analysis

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Abstract

Expressing emotions on social media is becoming popular by the day. Several studies have documented the relation between how people express their emotions to their mental state e.g., depression. Most of these studies were conducted in English, however, understanding how emotions are expressed using Arabic language has yet to be understood. This paper investigates how Arab Twitter users perceive depression on social media and analyze their sentiments about it. In addition, examining how Arabs express negative emotions in Arabic texts on social media platforms. A corpus analysis approach was used to understand how Arab users are expressing their emotions. In addition, a sentiment analysis was conducted to evaluate the intensity of negative feelings used in Twitter. Results showed a common theme exists when talking about depression and fear, Allah (God), and the usage of first pronouns. There is also a correlation between religion and depression when expressing emotions. It is evidently clear from the findings that there is a direct relation between how Arab users express their emotions and their perception of depression. One major implication was the nature of Arabic language, as there are different dialects of Arabic and one specific feeling, such as fear or sadness, can be expressed using numerous words and phrases.

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Keywords: Depression; Sentiment Analysis; Emotions; Social media; Arab

1. Introduction

Mental illness is a worldwide health problem affecting thousands of people and causing numerous deaths yearly. Depression is considered the leading cause of disability worldwide [1]. Individuals suffering from depression show

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symptoms in various ways, whether verbal or behavioral, which can manifest in changes in their online activity, like posting on social media platforms when expressing their emotions [2].

Social media is "a new fertile source" [3] for mental health observation. Users tend to favor text to reveal their feelings more than audio or video recordings. Zaghouani believes that social media platforms have become a critical online resource for understanding and studying the language used to express matters related to mental health, such as depression [4]. Sentiment analysis of social media texts is a crucial research method in healthcare and can provide essential insights [3].

Vedula & Parthasarathy (2017) studied social media interactions to identify signs of depression, observing reduced online activity, increased negative emotions, and distinct linguistic styles. They developed a classifier predicting depression with a 90% F-score, supporting the idea that social media can be a useful diagnostic tool [5]. Similarly, Ortega-Mendoza et al. (2022) found that analyzing the use of singular first-person pronouns in communication helps predict depression, showing that the context of pronouns like "I" or "you" is key in identifying depressive symptoms. This approach enhances the detection of depression [6]. Additionally, Iglesias & Moreno (2019) noted that advancements in sentiment analysis have enabled its application in health fields, with recent technological developments addressing traditional limitations caused by insufficient linguistic resources [7].

However, most of the studies have been conducted using English language datasets, and not enough studies have been carried out using Arabic data. This paper investigates how Arab Twitter users perceive depression on social media and analyze their sentiments. In addition, it will examine how Arabs express negative emotions using posts on social media platforms.

2. Methodology

The methodology proposed in this research involves four steps. In Step 1, we collected words related to depression and categorized them based on their relevance to depression and normal conversation. For Step 2, we imported the lexicon into the TAGS tool, grouped the results into an Excel sheet, and filtered out repetitive tweets. Step 3 involved selecting a sample of 1700 tweets, establishing annotation guidelines and choosing annotators for manual annotation. Finally, in Step 4, we utilized the Voyant tool for analysis.

2.1. Building the lexicon

The Atlas of Emotions developed by Paul Ekman, a leading researcher on emotions, was used as a foundation to build the lexicon. Ekman categorizes emotions into five main groups, four of which are related to depression: anger, fear, sadness, and disgust [8]. These feelings can be expressed using various forms of words and phrases, some of which were collected from similar previous studies [9], and also medical references such as the list of symptoms and feelings by Healthline website [8]. A total of 120 words were collected and classified into two categories: 40 words referring to both normal and depression-related conversations, and 80 words exclusively related to depression. The latter category was further divided based on depression-related emotions, as illustrated in Table 1.

Table 1. Negative Words Used in Part Two of the Lexicon

Emotions	Words and phrases
Loneliness	وحيدة، وحيد، منطوي، منطوية، انطواء، انعزال
Pessimism	تشاؤم، متشائم، متشائمة، فاقد الأمل، فاقدة الأمل، يأس، يأست
Sadness	حزينة، حزين، حزن
Anger	معصبة، معصب، عصبي، عصبية، متنرفز، متنرفزة، نرفزة
Depression	مكتئبة، اكتئاب، كآبة
Stress	متوتر، متوترة
Irritation	متضايقة، متضايق، مضايقة، مضايق، ضايق، ضايقة
Moodiness	متنكد، متنكدة، نكد، نكدي، نكدية، متكدرة، متكدر، كدر، منفسه، منفس

Fear	أخاف، خايف، خايفة، أخاف، خوف، مرعوبة، مرعوب، رعب
Worry	مهمومة، مهموم، قلقان، قلقانة
Doubts	شاك، شك، شكوك، شاكة
Lost	مشتتة، مشتت، تشتيت، تايه، تايهه
Suicide	انتحر، انتحار
Medications	بروزاك، زنكس، فلوكستين، زولفت
Medical terms	دكتور نفسي، طبيب نفسي، مرشد نفسي، علاج نفسي، أدوية نفسية، مريض نفسي، مريض اكتئاب، انكزايتي

2.2. Building the Corpus

A corpus is "a collection of machine-readable authentic texts…sampled to be representative of a particular language variety" [10]. It is a valuable tool when investigating matters or questions related to language usage [11]. For this paper, a corpus was built using the previously constructed lexicon. The 80 words in part two of the lexicon, which exclusively refer to depression and negative emotions, were imported to the TAGS tool developed by Martin Hawksey [12]. It is a "free Google sheet template" that enables individuals to run automated search results from the platform [13]. Since the TAGS tool had a maximum number of words inserted in one query for research, part two was divided into several Google sheets. The tool ran for a period of seven to ten days in the first quarter of 2022. All sheets were later combined into one Excel document. Tweets were then filtered using Microsoft Excel to select and delete repetitive tweets that were then sorted based on followers count from highest to lowest using the same software. The total number of tweets exceeded 200,000 tweets.

2.3. Annotation

The follower count was used to indicate how influential and impactful the Twitter account. The top 1700 tweets with the highest followers count were chosen as annotation samples. Table 2 demonstrates an example of the annotation samples. Then the annotation of the hate speech project in Micromappers website was used to obtain questions for the annotation. Four questions were chosen; what the Tweet Writer's Emotions are, how strong is the emotion in a tweet, what is the mental or psychological state in the tweet, and is the tweet positive or negative? A list of possible answers was provided for each question. These four questions will help the researcher understand how Twitter users express different feelings, assess its strength to determine the extent of significance, and analyze its influence. Further filtration was needed in this stage, as there were common words between the Arabic and Persian languages, and all non-Arabic tweets were deleted. Several annotators were selected for this stage of the research based on their performance in a sample task designed to assess their understanding of the annotation process. This selection ensured high-quality annotations and accurate assessments of each tweet, following the initial guidelines provided to them.

Table 2. Sample of the Annotation Samples.

English Translation	Text	User Followers Count
French Foreign Minister: The situation in Ukraine is very tense and united in the face of Russia	وزير خارجية فرنسا: الوضع في أوكرانيا متوتر للغاية ومتحدون في مواجهة روسيا #العربية_عاجل	21,193,791
Review by @sayish The movie #Friends_Will_Dear on Netflix.. a success by all standards, but I'm afraid of this "trend."	رأي بقلم @sayish فيلم #أصحاب_ولا_أعز على نتفليكس.. نجاح بكل المقاييس، ولكني أخاف من هذا "الترند"	12,824,304
And as the heroine runs in the cinema, she is afraid not to die..But I am standing in my place from my fear to live !Shall I tell you my fear?	وبتجري البطلة في السيما خايفة لا تموت.. إنما أنا واقف في مكاني من خوفي لاعيش! أحكيلك خوفي؟	7,946,553

3. Analysis and Discussion

3.1. Voyant Tool

3.1.1. Corpus Overview (summary)

The corpus has a total of 6,193,748 words and 332,892 unique word forms. The Vocabulary density, the ratio of words to unique words in the document [14] is 0.054, and the average number of words per sentence is 45.7.

3.1.2. Themes

3.1.2.1. Terms

This feature is essential for performing a discourse analysis of a mass of linguistic information. Table 3 below presents a list of the most frequent ten words in the corpus. These words "best define" a text or group of texts. The trends measurement reflects the distribution of each term within different parts of the corpus. The most dominant theme is fear, with a total count of (123,983). Four out of the ten most frequent words, ("Fear")خوف, ("Afraid")اخاف, ("Afraid")خايف, and ("Afraid")أخاف, are related to fear. Another common theme is Allah, the name of God among Muslims, with a count of (33,527). The use of first-person pronouns such as (Myself) نفسي, (I) أنا, and (Mine) لي, with a total count of (37,270), was also prominent.

It is essential to highlight that نفسي can also mean (psychological) in Arabic; however, in contexts related to psychology, the word نفسي(psychological) must be added to or combined with another word to indicate that meaning. For illustration, نفسيalone means (Myself), نفسيcombined with دكتور(doctor) means دكتور نفسي (therapist).

Table 3. Most Frequent Ten Words

#	1	2	3	4	5	6	7	8	9	10
Term	خوف ("Fear")	الله ("Allah")	اخاف ("Afraid")	نفسي ("Myself," "Psychological")	خايف ("Afraid")	أخاف ("Afraid")	بس ("But")	أنا ("I")	يا ("Oh")	لي ("mine")
Frequency	45754	33527	27309	27094	26641	24279	23539	22102	15377	15168

3.1.2.2. cirrus features

The cirrus feature visualizes the most frequent words of a corpus. The larger the word appears in the graphic, the more frequently it appears in the text. Figure 1 illustrates a word cloud of the 55 most frequent words in the corpus. The word اكتئاب ("depression") is present, as are other words that indicate negative feelings. The word الله ("God") is also present along with other related religious words like اللهم("making prayers"). Some words can reflect the theme of life, for example, الأرض("earth"), الناس ("people"), الدنيا ("life"), شخص ("person"), and العالمي("universal").

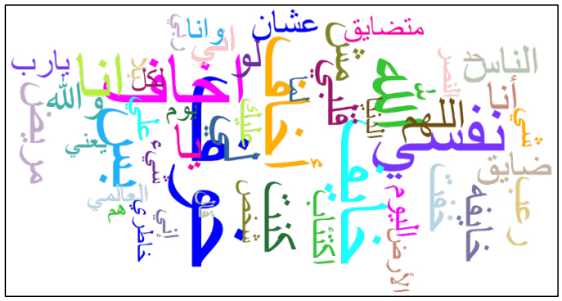

Figure 1. word cloud of the most 55 frequent words

3.1.3. Correlations

The correlations tool helps explore the extent of co-variation of term frequencies, whether they increase or decrease together or inversely that was obtained from the Voyant tool. For every two terms, Pearson's correlation coefficient r is calculated. The significance value is a measure of confidence in the correlation value. In general, a significance of

.05 or less shows a strong correlation. The correlation of three words was explored: “depression,” “Allah,” and “afraid.” Correlation values are sorted from largest to smallest.

Table 4 shows the correlations of the word “depression.” Overall, its correlations are positive except for one word, أمن(“safety”). The highest correlation is with the phrase أعبر (“express”). Other correlations can be sorted into two categories – feelings and actions. The correlation between depression and expressing negative emotions aligns with the existing literature. Scholars [15] indicated that behaviors of expressing negative feelings could intensify because of depression. One interesting finding is the correlation between depression and أسلحة(“weapons”) 0.831. The correlation between the two is directly proportional. According to a recent study, exposure to violence, particularly witnessing a shooting, increases the odds of depression [16].

Table 4. Correlations of the word “depression”

#	Term 1	← →	Term 2	Correlation (*r*)	Significance (*p*)
1	اعبر(“express”)		*اكتئاب (depression)	0.869	0.00111
2	أحزان(“sorrows”)		*اكتئاب (depression)	0.868	0.00113
3	افتكرت (“I remembered”)		*اكتئاب (depression)	0.866	0.00121
4	اصبرو(“be patient”)		*اكتئاب (depression)	0.844	0.00215
5	اسكتو (“shut up”)		*اكتئاب (depression)	0.843	0.00219
6	اتشافى (“healing”)		*اكتئاب (depression)	0.834	0.00268
7	اتنكدت (“felt bad”)		*اكتئاب (depression)	0.834	0.00268
8	اتكلمت (“talked”)		*اكتئاب (depression)	0.834	0.00273
9	أسلحة(“weapons”)		*اكتئاب (depression)	0.831	0.00291
10	أتحمل (“bear”)		*اكتئاب (depression)	0.825	0.00331
11	اجتاز(“overcome”)		*اكتئاب (depression)	0.824	0.00339
12	اذكر الله (“pray to Allah”)		*اكتئاب (depression)	0.815	0.00410
13	أمن (“safety”)		*اكتئاب (depression)	-0.843	0.00218

Table 5 demonstrates the correlation between the word “Allah” with both negative and positive feelings. The only two positive feelings are أحبك (“I love you”) and الأمان (“safety”). In contrast, most of the words reflected negative emotions yet showed differences in the type of correlation, whether positive or negative. The variance of emotions indicates a strong relationship with Allah.

Table 5. Correlations of the word “Allah”

#	Term 1	← →	Term 2	Correlation (*r*)	Significance (*p*)
1	اتغلب (overcome)		الله (Allah)	0.85744	0.001516
2	أحبك (I love you)		الله (Allah)	0.839257	0.002394
3	الخطر(danger)		الله (Allah)	0.819999	0.003673
4	أخاف (afraid)		الله (Allah)	0.803755	0.005082
5	أنتحر (suicide)		الله (Allah)	0.800939	0.005361
6	الأمان (safety)		الله (Allah)	0.788254	0.006751
7	الخوف (fear)		الله (Allah)	0.772602	0.008799
8	(my sorrows)ألمي		الله (Allah)	0.768605	0.009385
9	أتألم (in pain)		الله (Allah)	0.767953	0.009483
10	الالشيء(emptiness)		الله (Allah)	-0.76766	0.009528

Table 6 illustrates correlations of the of the word “afraid.” The majority of the correlations are negative. The only three words that correlate positively are أموت(“I die”), اندم(“I regret”), انتحار(“suicide”), and أحزن(“feel sad”). The higher the fear, the higher the negative feelings of death, regret suicide, and sadness. Some words reflected positive meanings, such as أنجح(“I succeed”) and اجتهاد(“hard work”).

Table 6. Correlations of synonyms of afraid

#	Term 1	←	→	Term 2	Correlation (r)	Significance (p)
1	أموت(“ I die”)			اخاف(afraid)	0.864999	0.001231
2	اندم(“I regret”)			خايف(afraid)	0.856253	0.001565
3	أنتحآر(“suicide”)			اخاف(afraid)	0.814416	0.004121
4	أحزن(“feel sad”)			اخاف(afraid)	0.803897	0.005069
5	اتضايق(“feel bad”)			اخاف(afraid)	-0.78851	0.006721
6	إكتئاب(“depression”)			اخاف(afraid)	-0.79105	0.006425
7	أنجح(“succeed”)			اخاف(afraid)	-0.79863	0.005597
8	اتجاهل(“ignore”)			اخاف(afraid)	-0.80286	0.00517
9	اجتهاد(“work hard”)			اخاف(afraid)	-0.8191	0.003743
10	اجهل(“don’t know”)			اخاف(afraid)	-0.85483	0.001625

3.1.4. Collocates

The collocates are the terms that appear most frequently in proximity to keywords across the entire corpus. The collocates of the corpus keywords are shown in Table 7. Most of the collocations are linked to the word “fear.” Another keyword is the word الله(“Allah”), which collocates mainly with the two words الله (Allah) (5761) and أخاف (“I am afraid”) (3412). It is suggested that the collocation between the two matching words الله (Allah) is due to the high frequency of the word in the corpus. The rest of the collocates were related to first-person pronouns, such as the words أمري (“my life”) (3228) and أبوابي (“my doors”) (3185). Which aligns with a similar study conducted in English Language by Bathina et al. (2021) confirming that language usage changes among depressed individuals, explicitly using more frequent terms that express negative emotions and first-person pronouns [17]. In Islam, these collocates are common forms of prayer (*Du'ā*) (du’a) to Allah. The results reflect a strong link between fear and religion.

Table 7. Collocates of top keywords

#	Term	Collocate	Count (context)
1	خوف(“fear”)	قلبي(“my heart”)	6635
2	هللا(“Allah”)	هللا(“Allah”)	5761
3	أخاف(“afraid”)	هللا(“Allah”)	3648
4	خوف(“fear”)	ضعف(“weakness”)	3624
5	هللا(“Allah”)	أخاف(“afraid”)	3412
6	خوف(“fear”)	يكسرني(“breaks me”)	3287
7	لي(“mine”)	أمري(“my life)	3228
8	لي(“mine”)	أبوابي(“my doors)	3185
9	خوف(“fear”)	وأزح(“remove)	3177
10	خوف(“fear”)	يسكنني(“inside me)	3157

3.1.5. Contexts

The Contexts, or “keywords,” in context features demonstrate each keyword occurrence surrounded by text. This feature helps understand how words are used in different contexts. We considered the context of the following three keywords: “depression,” “fear,” and “Allah.” The total window of words chosen for each keyword is five. Table 8 shows the contexts of the word “depression.” The contexts reflected multiple topics or themes, such as types of depression such as postpartum depression [18]. Also, signs and symptoms of depression include wasting time and lack of motivation [19]. Other topics discussed possible treatments for depression, like changing sleeping patterns.

Table 8. Contexts of the word depression

left	Term	Right
"ب ال أسباب "غير نظام نومك Change your sleeping pattern 💔😔	اكتئاب Depression اكتئاب Depression	إذا أصابك هلع خوف رهبة If you panic or feel scared مرحلة مابعد التخرج دي بتعمل

بعد الحمل وتخن وغير اسمه Postpartum depression and changed his name	Depressionاكتئاب	Post-graduation period causes التجميل والكالم ده بس جالهgot he But
اني ضيعت اليوم:سبب االكتئاب Depression cause is wasting time	Depressionاكتئاب	سبب تضييع اليوم The reason for wasting time
الدنيا مابي اداوم I don't want to go to work	Depressionاكتئاب	اخخخ فيني feel I Ooh
و لقيت أن عندك منهم If you have any	Depressionاكتئاب	حاجات تدول انك عندك5 Five signs indicated you have
إلى تحطيم اكثر فاكثر To more and more breaking	Depressionاكتئاب	الإنتاجية والانجاز و تنتقل من Productivity and achievements and get

In Table 9, the contexts of the word "Allah" are illustrated. Overall, prayer (*Du'ā*) is the dominant topic where individuals ask Allah for strength, safety, and protection. There was also mention of a verse from the Holy Quran. Although some of the contexts reflected that depression is an illness, other contexts suggest that users believe that faith, being close to Allah, is a treatment for depression, and being away from Allah can eventually lead to suicide. This aligns with the findings of earlier studies that highlighted that people with high levels of spirituality and religious involvement are at a lower risk of depression than those with no religious affiliation. It also suggested that some religious involvement can protect individuals against developing depression [20], [21].

Table 9. Contexts of the word "Allah"

left	Term	Right
ق ّوة واكتفاء وإغاثة، و مدد ال Strength and help	اللهAllah	التوكل على "ق ّوة إال باهلل Trust
لي ولك ولكل المتابعين السالمه For you and me and everyone's health	اللهAllah	ايضاً وال يستهان به نسأل Do not underestimate it. Ask
االمثال Examples	اللهAllah	يضرب) :واذكر السبب لآليات االتيه Mention the reason for these qur'anic versus
يعين لو حظوظ النفس، نلمح aid us	اللهAllah	دام..ماعلينا خوف We will not feel scared as
يبعده عنا وعنكم وال يمكن Keep it away from us, and you	الله Allah	االكتئاب مرض.الخ مش بمسلسل Not a TV series. Depression is illness
قيرل باي Girl bye	اللهAllah	انتحار اكثر عشنهم بعاد عن More suicide because they are away from
.. فهو صح يحميك من االكتئاب from you protect will Truly depression	اللهAllah	ثانيا بالنسبة للقرآن والتقرب من Second, regarding Quran and being close to
أني أخاف أن أؤذي أحد I am afraid to hurt anyone	اللهAllah	فليشهد Testify

Table 10 shows the contexts of the word "fear." Multiple themes were illustrated, while the theme of religion dominated. Fear was associated with praying (Du'a) and asking Allah for strength. Also, the word "fear" was associated with change, rejection, confrontation, and continuity.

Table 10. Contexts of the word "fear"

Left	Term	Right
يالزمك مثل ظلك ويشعرك أنك	خوف Fear	بأنك مريض وأنت لست بمريض

With you as if it's your shadow من المواجهة وهؤلاء بحاجة لتعامل From confrontation and those need يسكنني وكل ضعف يكسرني وكل Inside me, every weakness that breaks me من/ من الوحدة /من الرفض Of rejection, loneliness, of من بشر و ال رعب of human or terror	خوف Fear خوف fear خوف Fear خوف fear	You are sick but you are not sick with تعاملك ضعف وحكمتك في التجاهل ignoring in is wisdom Your قلبي وأزح من قلبي كل And take from my heart all خوف مع تنوع انواعة #مصدرها!! types all of Fear مع هللا ال.ما يملك With Allah there is no
تذكر.من التغيير الي سيحدث Of change to what will happen من بشر و ال رعب of human or terror	خوف fear خوف Fear	في حياتك ولكن تجيك مشاعر feel you but life your In الغد يقلقني فاهلل يكفيني مع هللا ال Allah is enough for me, with Allah there is no

3.2. Sentiment Analysis

3.2.1. Overview

The sentiment analysis was conducted on the top 1700 tweets collected using TAGS V.6.1 tool. It was evaluated how often the list of feelings was identified for question one, as shown in Figure 2. The results show that the reference to محايد("neutral") is the most dominant, based on the annotation results with a percentage of around 22%. This finding was expected as the top tweets were selected from accounts with the highest number of followers, accounts of news channels. Even though they included expressions with negative words such as خائف("scared"), the dominant feeling of these tweets was neutral due to the structure of journalistic writing and language usage [22], [23].

Another feeling commonly referred to in the tweets is خوف("fear"), with 13% of the overall tweets evaluated. Another feeling observed was تفائل("hope"), with around 11.7%. The difference between the frequency of references to these two feelings of fear and hope is 1.3%. These results can be attributed to the strong connection between faith and hope in Islam, as suggested by multiple scholars [21], [24], [25].

Lastly, حزين("sad") was referenced by 9.4% of the Twitter users. This value is significant as it indicates the number of tweets with negative emotions.

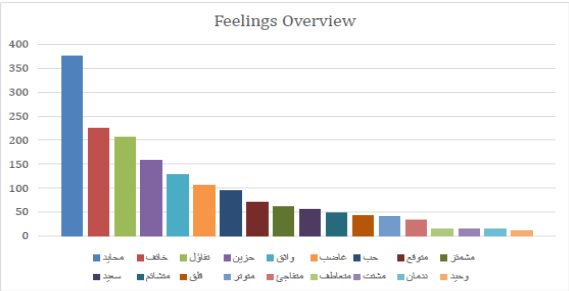

Figure 2. feelings: Overview

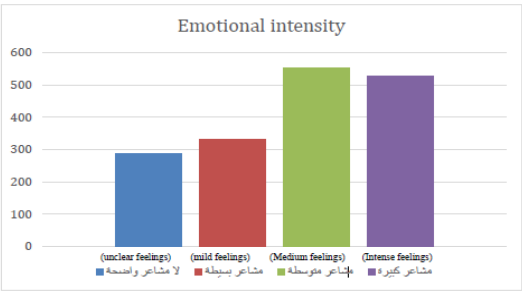

Figure 3. Emotional Intensity

3.2.1.1. Emotional Intensity

The study considers four levels of emotional intensity. These levels were introduced in the list of possible answers to the second question in the annotation sample. The levels are unclear, mild, medium, and intense. Figure 3 shows that most tweets demonstrate "strong" and "medium" intensity of emotions, with a combined percentage of 63.5%. This finding can be attributed to how social media is used to express emotions, which validates the point of selecting Twitter as a platform that can be used to evaluate the intensity of emotions. This finding aligns with multiple scholars in the

field [7], [26], [27]. Low-intensity emotions were around 19.5%. Lastly, the unclear emotions were observed to be 17%. The “low” and “unclear” emotion intensity results can be attributed to the number of tweets related to news, where the dominant emotions were neutral.

3.2.1.2. Detecting Valence

Valence is a crucial feature of emotions, and it refers to “the positive or negative charge associated with a particular physical or mental state, or a particular combination of these” [28]. It is often described using polar scales that define a dimension ranging from pleasantness, such as being happy and hopeful, to unpleasantness, such as a state of unhappiness or annoyance. The valence dimension exists in almost all systems built to categorize emotion [29], and certain situations often trigger contrasting emotions. For example, undergoing a new experience like riding a huge roller coaster could be terrifying (negative valence), yet individuals may still feel happy and excited about the experience (positive valence) [30]. Figure 4 shows the valence detection of the annotated sample. It was found that the highest valence was the “neutral” or “mix,” which can be attributed to the high number of tweets related to news shared by Twitter accounts of news channels. Another reason is that some tweets were inconclusive when evaluated, so it was impossible to determine their valence, so the nearest conclusion was neutral or a “mix of hedonic tones.” The second highest valence was negative to some extent, with 25.17% of the annotated tweets, followed by 17.64% for the “positive to some extent” valence. The negative emotion was slightly higher than the extreme positive for extreme valences by 2.6%. This suggests a similar qualitative evaluation for positive and negative impacts.

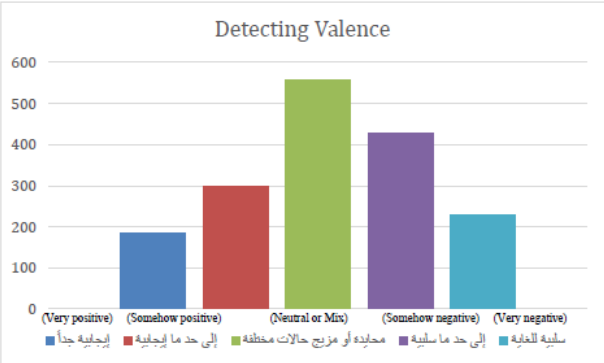

Figure 4. Detecting Valence

4. Conclusion

In conclusion, this study has investigated perceptions of depression among Arab social media users and how negative feelings and emotions are expressed on social media platforms. Although there is a lot of research on the increasing role of social media data in detecting and understanding depression, there are not enough studies examining data from Arab social media users. Chapter Two discussed the methodology of the study: a general overview of the research approach, stages of building the lexicon, stages of building the corpus, and the manual annotation process. Next, the text analysis tool, Voyant, was utilized to explore the topic in-depth, using features such as contexts, word cloud, collocates, and keywords. The analysis suggests that the common themes when talking about depression are fear and Allah. Another significant finding was the high prominence of first-person pronouns usage. There is also a correlation between religion and depression when expressing emotions. The manual annotation results validated that people said their feelings on Twitter with different intensities. These findings are significant in establishing a foundation for future research on detecting depression among Arab social media users using the same approach presented in this study. These research findings demonstrate that the exploratory has answered the main research question: it has reflected how depression is perceived and expressed among Arab social media users. Future work might entail using a corpus of texts written exclusively by Arab individuals that suffer from depression would add significant information about language usage and expressing emotions.

References

[1] A. Krygsman, A. H. Farrell, H. Brittain, and T. Vaillancourt, 'Depression Symptoms, Mattering, and Anti-mattering: Longitudinal Associations in Young Adulthood', *Journal of Psychoeducational Assessment*, vol. 40, no. 1, Art. no. 1, Feb. 2022, doi: 10.1177/07342829211050519.

[2] H. Zogan, I. Razzak, X. Wang, S. Jameel, and G. Xu, 'Explainable depression detection with multi-aspect features using a hybrid deep learning model on social media', *World Wide Web*, Jan. 2022, doi: 10.1007/s11280-021-00992-2.

[3] S. Almouzini, M. khemakhem, and A. Alageel, 'Detecting Arabic Depressed Users from Twitter Data', *Procedia Computer Science*, vol. 163, pp. 257–265, Jan. 2019, doi: 10.1016/j.procs.2019.12.107.

[4] W. Zaghouani, 'A Large-Scale Social Media Corpus for the Detection of Youth Depression (Project Note)', *Procedia Computer Science*, vol. 142, pp. 347–351, Jan. 2018, doi: 10.1016/j.procs.2018.10.483.

[5] N. Vedula and S. Parthasarathy, 'Emotional and Linguistic Cues of Depression from Social Media', in *Proceedings of the 2017 International Conference on Digital Health*, in DH '17. New York, NY, USA: Association for Computing Machinery, Jul. 2017, pp. 127–136. doi: 10.1145/3079452.3079465.

[6] R. M. Ortega-Mendoza, D. I. Hernández-Farías, M. Montes-y-Gómez, and L. Villaseñor-Pineda, 'Revealing traces of depression through personal statements analysis in social media', *Artificial Intelligence in Medicine*, vol. 123, p. 102202, Jan. 2022, doi: 10.1016/j.artmed.2021.102202.

[7] C. A. Iglesias and A. Moreno, 'Sentiment Analysis for Social Media', *Applied Sciences*, vol. 9, no. 23, Art. no. 23, 2019, doi: 10.3390/app9235037.

[8] C. Raypole, 'List of Emotions: 54 Ways to Say What You're Feeling', Healthline. Accessed: Feb. 12, 2022. [Online]. Available: https://www.healthline.com/health/list-of-emotions

[9] R. M. Ortega-Mendoza, D. I. Hernández-Farías, M. Montes-y-Gómez, and L. Villaseñor-Pineda, 'Revealing traces of depression through personal statements analysis in social media', *Artificial Intelligence in Medicine*, vol. 123, p. 102202, Jan. 2022, doi: 10.1016/j.artmed.2021.102202.

[10] T. McEnery, R. Xiao, and Y. Tono, *Corpus-based Language Studies: An Advanced Resource Book*. Taylor & Francis, 2006.

[11] R. Reppen, 'Building a corpus: what are key considerations?', in *The Routledge Handbook of Corpus Linguistics*, 2nd ed., Routledge, 2022.

[12] 'TAGS', TAGS. Accessed: Mar. 11, 2022. [Online]. Available: https://tags.hawksey.info/

[13] E. H. Alkhammash, J. Jussila, M. D. Lytras, and A. Visvizi, 'Annotation of Smart Cities Twitter Micro-Contents for Enhanced Citizen's Engagement', *IEEE Access*, vol. 7, pp. 116267–116276, 2019, doi: 10.1109/ACCESS.2019.2935186.

[14] SUEISHAQLAB, 'Finding the write words', The Ishaq Lab. Accessed: Mar. 25, 2024. [Online]. Available: https://sueishaqlab.org/2017/12/07/finding-the-write-words/

[15] J. Rottenberg and C. Vaughan, 'Emotion Expression in Depression: Emerging Evidence forEmotion Context-Insensitivity', in *Emotion Regulation: Conceptual and Clinical Issues*, A. J. J. M. Vingerhoets, I. Nyklíček, and J. Denollet, Eds., Boston, MA: Springer US, 2008, pp. 125–139. doi: 10.1007/978-0-387-29986-0_8.

[16] C. Leibbrand, F. Rivara, and A. Rowhani-Rahbar, 'Gun Violence Exposure and Experiences of Depression Among Mothers', *Prevention Science*, vol. 22, no. 4, pp. 523–533, May 2021, doi: 10.1007/s11121-020-01202-7.

[17] K. C. Bathina, M. ten Thij, L. Lorenzo-Luaces, L. A. Rutter, and J. Bollen, 'Individuals with depression express more distorted thinking on social media', *Nat Hum Behav*, vol. 5, no. 4, pp. 458–466, Apr. 2021, doi: 10.1038/s41562-021-01050-7.

[18] J. Carberg, 'Statistics on Postpartum Depression - Postpartum Depression Resources', PostpartumDepression.org. Accessed: Mar. 07, 2022. [Online]. Available: https://www.postpartumdepression.org/resources/statistics/

[19] B. Smith, 'Depression and motivation', *Phenomenology and the Cognitive Sciences*, vol. 12, no. 4, Art. no. 4, Dec. 2013, doi: 10.1007/s11097-012-9264-0.

[20] H. G. Koenig, 'Religion and Depression in Older Medical Inpatients', *The American Journal of Geriatric Psychiatry*, vol. 15, no. 4, Art. no. 4, Apr. 2007, doi: 10.1097/01.JGP.0000246875.93674.0c.

[21] H. G. Koenig and D. B. Larson, 'Religion and mental health: evidence for an association', *null*, vol. 13, no. 2, Art. no. 2, Jan. 2001, doi: 10.1080/09540260124661.

[22] M. Müller, 'Spreading the word? European Union agencies and social media attention', *Government Information Quarterly*, p. 101682, 2022.

[23] K. Welbers and M. Opgenhaffen, 'Presenting news on social media: Media logic in the communication style of newspapers on Facebook', *Digital Journalism*, vol. 7, no. 1, pp. 45–62, 2019.

[24] J. E. Hupcey, 'Feeling Safe: The Psychosocial Needs of ICU Patients', *Journal of Nursing Scholarship*, vol. 32, no. 4, Art. no. 4, Dec. 2000, doi: 10.1111/j.1547-5069.2000.00361.x.

[25] *Handbook of the psychology of religion and spirituality, 2nd ed.* in Handbook of the psychology of religion and spirituality, 2nd ed. New York, NY, US: The Guilford Press, 2013, pp. xvi, 698.

[26] D. Tanna, M. Dudhane, A. Sardar, K. Deshpande, and N. Deshmukh, 'Sentiment Analysis on Social Media for Emotion Classification', in *2020 4th International Conference on Intelligent Computing and Control Systems (ICICCS)*, May 2020, pp. 911–915. doi: 10.1109/ICICCS48265.2020.9121057.

[27] Z. Wang, C. S. Chong, L. Lan, Y. Yang, S. Beng Ho, and J. C. Tong, 'Fine-grained sentiment analysis of social media with emotion sensing', in *2016 Future Technologies Conference (FTC)*, Dec. 2016, pp. 1361–1364. doi: 10.1109/FTC.2016.7821783.

[28] L. C. Charland, 'Emotion Experience and the Indeterminacy of Valence.', in *Emotion and consciousness.*, New York, NY, US: The Guilford Press, 2005, pp. 231–254. doi: 10.1016/j.tics.2007.01.005.

[29] F. Citron, M. Gray, H. Critchley, B. Weekes, and E. Ferstl, 'Emotional valence and arousal affect reading in an interactive way: Neuroimaging evidence for an approach-withdrawal framework', *Neuropsychologia*, vol. 56, pp. 79–89, Apr. 2014, doi: 10.1016/j.neuropsychologia.2014.01.002.

[30] S. Hussain and S. Alhabash, 'Nostalgic Emotional Valence and Its Effects on Help-Seeking in Depression. An Application of the Theory of Planned Behavior', *Health Communication*, vol. 36, pp. 1–12, Jul. 2020, doi: 10.1080/10410236.2020.1794549.